

* DDL Statement

Questions:-

1. Add a column before the ^{name} name column.
2. Change the data type of column to integer data type.
3. Add a not null constraint to name and age column.
4. Add a foreign key with column depid.

* DML:-

```
SET SQL_SAFE_UPDATES = 0;
```

This query is use when MySQL Workbench or MySQL give error "Using safe update mode."

When use this command, then update the data through update command.

* Interview Questions:-

Q. What is DDL vs DML

1. Difference Truncate, delete and drop.

2. Normalization

3. Difference varchar and char datatypes.

* TCL → is used to manage transaction in a database.
Transaction → transaction are the group of sql statement which work as ~~single~~ unit.

This TCL command ensure the data is correct and accurate. make

• ~~Atomicity~~ → Command (i) COMMIT: - Save changes permanently.

UPDATE employee
 SET Salary = 6000000
 Where emp-id = 1;

• Consistency →

• ~~Start Transaction~~ →

COMMIT; → the changes are permanently stored ~~for~~.

(ii) ROLLBACK: - Undoes changes since the last commit.

~~Commit/RollBack/DPL Statement~~

UPDATE employee
 SET Salary = 6000000
 Where emp-id = 1;

ROLLBACK; → salary returns to its previous value.

(iii) SavePoint: → Creates a point within a transaction to which you can roll back.

UPDATE employee
 SET salary = 60000
 Where emp-id = 1;

SAVEPOINT sp1;

UPDATE employee
 SET salary = 70000
 Where emp-id = 1;

(iv) RollBack to Savepoint:
 Returns to a specific savepoint
 Rollback To sp1;

* Difference between Char and Varchar.

Char	Varchar
Stores fixed-length character data.	Store variable-length character data.
Use full allocated space.	Use only required space.
fixed length	Fixed Variable length
Add spaces if data is shorter than defined length.	Does not add extra spaces.
Ex: Char(10)	Varchar(10)

* DEL → Data Control language is used to control access and permissions in a database.
Select * from mysql.user;

Create user ishan identified by '123';

Select current user();

Create database testdb4;

Use testdb4;

Create table emp(id int, name varchar(20));

insert into emp values (20, 'abhi'), (21, 'name');

select * from emp;

ishan user
ishan password
123

Questions

- ① Revoke this delete privilege ishan user.
- ② Give select and insert " to a particular table.
- ③ Create a database with 3 columns and insert at least 2 row.
- ④ Remove all the privileges from ishan user.

Create 3 users and also create a ~~role~~ role.

* DCL Commands: - ① GRANT: - Gives permissions to a user.

GRANT Select on employee

To user 1; → # user 1 can read data from employee table.

②. REVOKE → Removes permission from a user.

REVOKE Select on employee

from user 1; → # user 1 can no longer read data from employee table.

* Give Multiple Permissions: -

GRANT SELECT, INSERT, UPDATE
on employee To user 1;

Now user 1 can read, update and insert data.